

DESIGN PROJECT REPORT

Semantic Communication for Connected Autonomous Vehicles: Evaluating a Graph-Based Encoder–Decoder over a Realistic VANET

CSE 4610 : Design Project

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List of Abbreviations

ADS	Autonomous Driving System
BEV	Bird's Eye View
CAV	Connected Autonomous Vehicle
CDL	Clustered Delay Line
GBSED	Graph-Based Semantic Encoder–Decoder
IPM	Inverse Perspective Mapping
LSTM	Long Short-Term Memory
MR-GCN	Multi-Relational Graph Convolutional Network
OFDM	Orthogonal Frequency-Division Multiplexing
PDR	Packet Delivery Ratio
SNR	Signal-to-Noise Ratio
TraCI	Traffic Control Interface
VANET	Vehicular Ad-hoc Network
WSM	Wave Short Message

Table 1: Marks Distribution Summary

CO No.	Course Outcome	Contributing Chapters	Marks
CO1	Identify research problems and implement innovative solutions by exploring, analyzing, and applying state-of-the-art technologies, frameworks, tools, and algorithms to address complex real-world engineering challenges.	Chapter 1, 2 and 3	10
CO2	Demonstrate effective teamwork, professional ethics, and responsibility by applying project management principles to plan, execute, and deliver a complex software system within defined constraints.	Appendix A and B	5
CO3	Evaluate and design computing solutions considering ethical, societal, and environmental implications, ensuring responsible and sustainable engineering practices.	Chapter 4, 5 and 7	10
CO5	Build a strong foundation for future research in the same or related domains.	Chapter 8 and 9	5

Abstract

Connected autonomous vehicles must share what they see, but transmitting camera frames over a vehicular radio link is prohibitively expensive. Semantic communication proposes sending the *meaning* of a scene instead of its pixels. The Graph-Based Semantic Encoder–Decoder (GBSED) framework realises this idea by converting a road image into a scene graph, compressing it, and reconstructing it at the receiver. Its original evaluation, however, used a simulated MIMO–OFDM physical layer in isolation: an abstract signal-to-noise sweep with no vehicle mobility, no medium access contention, and no packet loss.

This project ports GBSED onto a realistic vehicular network — OMNeT++, Veins and SUMO over IEEE 802.11p — and asks a question the original evaluation could not answer: at a *fixed bit budget*, is a scene graph actually better than an image? We answer it with a controlled experiment in which each image is re-encoded to the exact byte budget of its scene graph, so that identical packets cross an identical radio channel and only the representation differs. Under that control, the semantic payload reconstructs every delivered frame perfectly while the pixel payload recovers none of the 44 safety-critical relations. A channel-free rate–distortion sweep shows that images need roughly 96× more bandwidth to convey the same safety-relevant content. We further identify and correct three evaluation defects in the existing pipeline, and design a slice-aligned payload format that converts a lost packet from the loss of an entire frame into the loss of a single relation type, raising safety-relation recall under partial delivery from 0.50 to 1.00 at no additional cost in airtime.

Chapter 1

Introduction

1.1 Introductory Information

A connected autonomous vehicle (CAV) perceives its surroundings through cameras, radar and lidar, and then must decide how to act. Many of the most valuable decisions — braking for a vehicle that another car can see but this one cannot, merging safely, cooperating at an intersection — depend on information that a *neighbouring* vehicle holds. This is the premise of cooperative perception: vehicles exchange what they observe so that each can act on more than its own sensors reveal.

The obvious way to do this is to transmit the sensor data itself. A single 1280×720 camera frame occupies roughly 1.2 MB as a JPEG. A vehicular radio link based on IEEE 802.11p [4] offers a nominal 6 Mbit/s shared among every vehicle in range, and two vehicles moving in opposite directions may remain within radio range of each other for only a few tens of seconds. The arithmetic does not work.

Semantic communication offers a different framing. Shannon’s original theory [10] deliberately set aside the *meaning* of a message and optimised only the faithful reproduction of symbols. Semantic and task-oriented communication revisits that decision [3], [13]: if the receiver’s purpose is known in advance, the transmitter need only send what that purpose requires. For a vehicle whose task is risk assessment, the relevant content is not the texture of the road surface but the identity, position and relationships of the traffic participants.

1.2 Motivations and Scope

The Graph-Based Semantic Encoder–Decoder (GBSED) framework [8] instantiates this idea for CAVs. It converts a road image into a *scene graph* — nodes for the ego vehicle, lanes, road and detected actors; edges for spatial and semantic relations such as `isIn`, `toLeftOf` and `near_coll` — compresses that graph by discarding relation types absent from the scene, transmits it, and reconstructs it at the receiver for a downstream risk classifier. The reported compression against raw imagery is 99.9%.

The original evaluation transmits this payload over a simulated MIMO–OFDM physical layer with a 3GPP CDL channel [1], sweeping signal-to-noise ratio from 0 to 20 dB. That is a rigorous physical-layer study, but it is not a network. It contains no vehicle mobility, no medium access control, no packet framing, no contention with other traffic, and no notion of a vehicle passing out of radio range. It also compares against raw imagery, which is not the comparison a sceptic would demand: nobody proposes transmitting uncompressed pixels.

This project addresses both gaps. The scope is deliberately narrow and complete: take the published semantic encoder and decoder as given, place them either side of a realistic vehicular network, and measure what happens.

1.3 Problem Statement

We address three questions.

1. **Does the semantic chain survive a real network?** Packet loss in a VANET is bursty and correlated with distance, not a smooth function of SNR. Does the encode–transmit–decode chain remain lossless when the channel is a real 802.11p link with mobility?
2. **At a fixed bit budget, is a scene graph better than an image?** A compression ratio against *raw* pixels is not a fair benchmark. The honest question is what a modern image codec achieves when given exactly the same number of bytes.
3. **Can the representation degrade gracefully?** A scene graph decomposes into independent relation slices. Can that structure be exploited so that losing a packet costs a relation type rather than the whole frame?

1.4 Research Challenges

Four difficulties shaped the work.

A fair comparison is hard to construct. Comparing a semantic payload against an image payload confounds two variables at once: the number of bytes and the meaning of those bytes. Any difference in delivery rate makes the comparison uninterpretable. The experiment had to be designed so that both arms produce byte-for-byte identical traffic patterns.

The scene graph representation is positional and fragile. Node names are never transmitted; they are re-derived at the receiver from label indices and a positional heuristic. Feature columns are identified by position in a Python dictionary. A single reordering silently corrupts every decoded graph without raising an error.

The existing evaluation metrics did not discriminate. As documented in Chapter 6, three of the four fidelity metrics in the inherited pipeline were the same number, and a fourth had a floor of 0.38 that a receiver could reach while understanding nothing at all.

The simulation had to be built before it could be run. Veins, OMNeT++ and SUMO are version-sensitive; the toolchain had to be assembled and debugged on two different operating systems before a single measurement was possible.

1.5 Contribution

1. **A working GBSED-VANET integration.** The published semantic layer operates across a full OMNeT++/Veins/SUMO simulation over IEEE 802.11p, with mobility-driven packet loss. The chain is lossless: every delivered frame reconstructs node- and edge-identically.
2. **A controlled, fixed-budget comparison against images.** Each frame's image is re-encoded to that frame's exact semantic byte budget, producing identical traffic. At that budget the pixel arm recovers 0 of 44 safety-critical relations against the semantic arm's 44.
3. **A rate-semantics curve** quantifying the gap: WebP first matches GBSED's safety-relation recall at 96× the bandwidth; JPEG does not reach it within 128×.
4. **A slice-aligned payload format** that makes partial delivery useful, raising safety-relation recall under loss from 0.50 to 1.00 at no extra airtime.

5. **Three corrected evaluation defects** — a metric that was algebraically identical to the delivery rate, a metric floor of 0.38 earned by transmitting nothing, and three channel configurations that were exact duplicates of three others.

1.6 Organization

Chapter 2 surveys scene-graph perception and semantic communication. Chapter 3 states the research gap precisely. Chapter 4 describes the methodology, and Chapter 5 the implementation. Chapter 6 presents and interprets the results. Chapter 7 considers ethical, social and environmental implications, Chapter 8 the limitations, and Chapter 9 concludes.

Chapter 2

Related Works

Three lines of work converge on this project: representing road scenes as graphs, using those graphs for risk assessment, and communicating meaning rather than bits.

2.1 From Pixels to Scene Graphs

Until recently, autonomous-vehicle perception relied either on convolutional networks operating directly on pixels or on model-based methods using known road geometry. Malawade et al. [6] argue that neither obtains a human-like understanding of a complex road scenario, because neither explicitly captures inter-object relationships or the overall structure of the scene. Cognitive studies suggest that humans identify the structure of a scene and reason about relations between objects when judging risk, which motivates an explicitly relational representation.

Their tool, `roadscene2vec`, extracts a scene graph from a video frame or from CARLA simulator output [2]. Objects are detected, projected into a bird’s eye view through inverse perspective mapping, and connected by relations determined by distance and bearing thresholds. The result is a directed multi-relational graph over a small, fixed ontology of actor types and relation types. This project uses `roadscene2vec` as its perception front end, and inherits both its strengths and its brittleness — in particular the fact that its relation and actor lists are *index spaces* whose ordering is a silent contract between encoder and decoder.

2.2 Scene Graphs for Risk Assessment

Yu et al. [14] demonstrate what the representation buys. They model the *subjective* risk of a driving manoeuvre as a supervised scene classification problem over sequences of scene graphs, using a multi-relational graph convolutional network [9] followed by an LSTM and attention layers. Their approach reaches 96.4% accuracy against 91.2% for a state-of-the-art pixel-based model on a large synthesised dataset, and the margin widens sharply on a small one (91.8% versus 71.2%). Most relevant here, a model trained on synthetic data transfers to real-world data at 87.8% accuracy against 70.3% for the pixel baseline.

Two conclusions matter for this project. First, the graph representation is *sufficient* for the risk task — the downstream model never needs the pixels. Second, it is more data-efficient and more transferable, which suggests the graph retains what matters and discards what does not. This is precisely the property a semantic communication system needs.

2.3 Semantic and Task-Oriented Communication

Shannon [10] explicitly excluded meaning from the engineering problem. Semantic communication revisits that exclusion. Gündüz et al. [3] frame the shift as moving beyond transmitting bits toward context- and task-aware transmission, and Xie et al. [13] demonstrate a learned end-to-end semantic system for text. Subsequent work spans generative models, vision transformers, and large language models used to produce compact transmissible descriptions.

Ribouh and Di Ngoma [8] bring this to vehicular networks with GBSED. Their semantic encoder maps an image to an adjacency tensor $T \in \{0, 1\}^{N \times N \times |R|}$ and a node feature matrix $F \in \mathbb{R}^{N \times d}$. Their semantic compression algorithm drops the adjacency matrices of relations absent from the scene, achieving an average 67% reduction of the tensor and 99.9% against raw imagery. The decoder reverses this and feeds a GNN–LSTM–MLP risk classifier. They report semantic fidelity above 0.9 for SNR greater than 10 dB and a downstream accuracy of 0.962.

2.4 Gaps and Opportunities

Taken together, these works establish that scene graphs are sufficient for risk assessment and that they can be transmitted compactly. Two gaps remain.

The channel is not a network. GBSED’s evaluation is a physical-layer study: an SNR sweep over a CDL channel, with the payload zero-padded to a fixed block size. Real vehicular communication adds mobility, medium access, framing, and range-dependent bursty loss. A system that performs well against smooth SNR degradation may behave quite differently when packets are simply absent.

The comparison baseline is too weak. Reporting 99.9% compression against *raw RGB* is technically correct but rhetorically unsatisfying: no deployed system transmits raw pixels. The paper does compare against JPEG, but only at JPEG’s own default quality — 5.29 GB for the dataset, a 2.47× reduction. The question it leaves open is what an image codec achieves when constrained to the *same byte budget* as the scene graph. That is the question this project answers.

Chapter 3

Problem Definition and Scope

3.1 Research Gaps

Gap 1 — Evaluation under an abstract channel. GBSED is validated against a MIMO-OFDM link with a CDL channel model. This captures fading and noise but not the network: no MAC contention, no mobility, no packet boundaries, and no loss pattern correlated with vehicle separation.

Gap 2 — An uncontrolled comparison. Compression ratios against raw or default-quality JPEG vary two things simultaneously — the number of bytes and what those bytes mean. Neither the semantic advantage nor the bandwidth advantage can be isolated.

Gap 3 — An unexploited architectural property. Semantic compression produces independent per-relation slices, and the decompression algorithm explicitly tolerates a partial relation set. Nothing in the existing pipeline uses this: the serialised payload is one flat vector, so any partial loss destroys the whole frame.

3.2 Problem Statement

Given a published semantic encoder f_{enc} and decoder f_{dec} operating on road scene graphs, and a realistic VANET simulation $\mathcal{N}$ comprising SUMO mobility and an IEEE 802.11p physical and MAC layer, we seek to determine:

1. the semantic fidelity of $f_{\text{dec}}(\mathcal{N}(f_{\text{enc}}(I)))$ as a function of channel condition and vehicle separation;

2. whether, for a payload budget B held exactly constant, a scene-graph representation preserves more task-relevant content than an image representation — and if so, the budget B' at which the image representation becomes competitive;
3. whether a payload layout aligned to relation-slice boundaries converts frame-level loss into relation-level degradation.

3.3 Scope and Delimitations

The downstream task is fixed in advance as sequence-level risk classification, which is what makes the comparison in question 2 meaningful. We do not retrain the perception or classification models; both are used as published. We evaluate two driving sequences from a lane-change dataset, and we do not claim statistical generality across dataset scale. The physical layer is IEEE 802.11p as modelled by Veins, not the MIMO-OFDM stack of the original work; the two are complementary evaluations rather than substitutes.

Chapter 4

Proposed Methodology

4.1 Overview of the Methodology

Figure 4.1 shows the complete pipeline. A road image is converted into a scene graph, the graph is encoded and compressed into a byte payload, the payload crosses a simulated 802.11p link between two moving vehicles, and the receiver reverses the process and runs the risk classifier.

A deliberate architectural decision underlies the whole design: **the semantic layer never executes inside the simulation**. Encoding is a pre-process, decoding a post-process, and the network moves opaque bytes between them. This is justified because the encoder’s output depends on nothing the simulation knows — the same image yields the same bytes regardless of where the vehicles are. Executing it in-loop would add approximately one second of wall-clock time per frame to a simulation that otherwise completes in 0.3 s, and would buy nothing. The exception, discussed in Chapter 9, is adaptive encoding, where the transmitter chooses what to send based on live link quality.

4.2 Chronological Breakdown of Components

4.2.1 Layer 1: Image to Scene Graph

An object detector localises road entities. The published implementation uses Detectron2’s Mask R-CNN, which requires CUDA; we substitute torchvision’s Faster R-CNN [7], which produces the same (boxes, labels, image size) triple and runs on CPU. Bounding boxes are projected into a bird’s eye view by inverse perspective mapping,

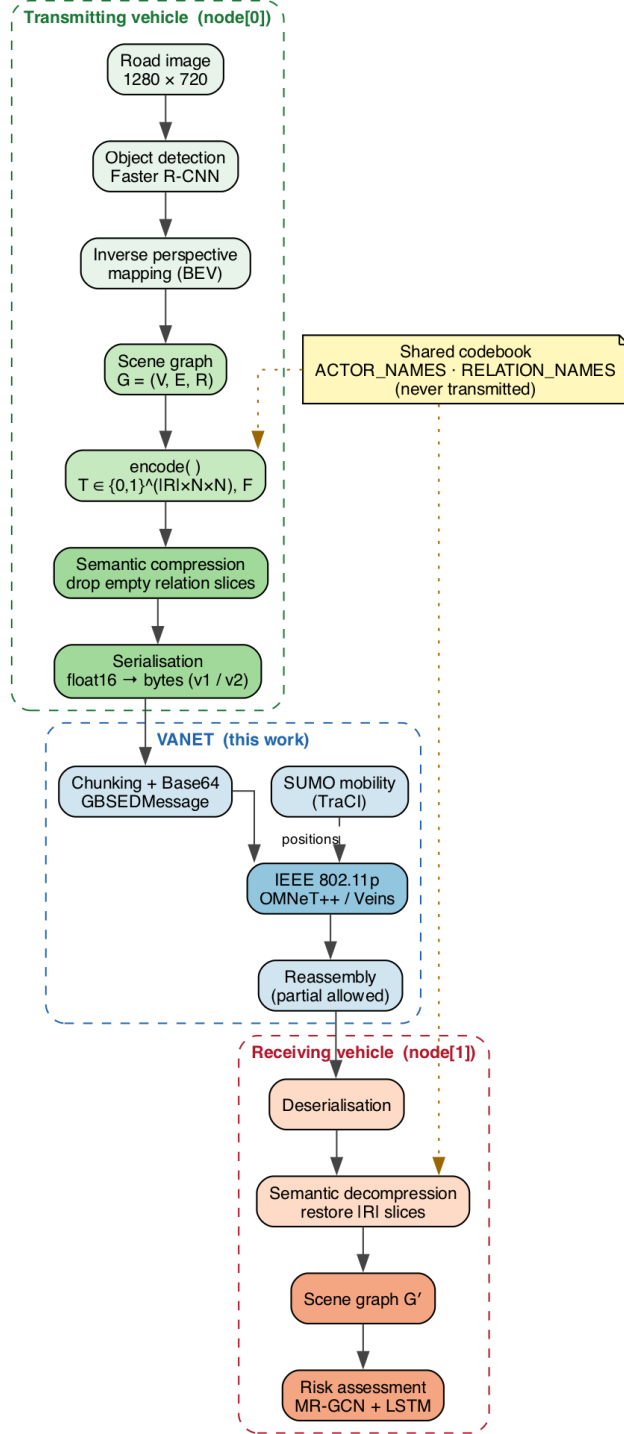


Figure 4.1: The GBSED semantic chain integrated with a VANET simulation. Green is the transmitting vehicle, blue the network contribution of this project, red the receiving vehicle. The shared codebook is never transmitted: node names are re-derived at the receiver from label indices.

and `roadscene2vec` constructs the graph: a fixed skeleton of a road node, an ego node and three lane nodes, plus one node per detected actor, connected by relations determined by distance and bearing thresholds.

4.2.2 Layer 2: Semantic Encoding and Compression

The graph is encoded as an adjacency tensor $T \in \{0, 1\}^{|R| \times N \times N}$ over $|R| = 12$ relation types and N nodes, plus a node feature matrix F . Semantic compression removes every relation slice that is entirely zero, yielding a compressed tensor T' and an index list L recording which relations survived. For the scenes in this study this discards between 4 and 8 of the 12 slices.

4.2.3 Layer 3: Serialisation

The arrays are flattened into a single `float16` vector, each section preceded by its length, and written as bytes. This flat serialisation is byte-for-byte identical to the published implementation.

4.2.4 Layer 4: Transport

The payload is split into fixed-size chunks, Base64-encoded, and broadcast as 802.11p Wave Short Messages. The receiver reassembles them.

4.2.5 Layer 5: Decoding and Task Inference

Deserialisation, semantic decompression and graph regeneration reverse the encoder. The reconstructed graph feeds an MR-GCN with an LSTM and attention layers for sequence-level risk classification.

4.3 The Fixed-Budget Comparison

The central methodological contribution is the design of a *controlled* comparison against a pixel baseline. Three arms are constructed:

Full image. The original JPEG, transmitted as-is. This answers a feasibility question.

Matched budget. Each image is re-encoded — by searching over scale and quality — to the largest representation that fits within *that frame’s exact GBSSED payload size*. Identical bytes produce an identical chunk count, hence identical packets

over an identical channel. Delivery is therefore identical by construction, and any difference in outcome is attributable to the representation alone.

Budget sweep. The budget is then multiplied by 1, 2, 4, ... , 128 and the measurement repeated without a channel, producing a rate–semantics curve.

Both arms run the same detector, the same graph extraction and the same scoring code. The received image is upscaled to 1280×720 before detection — not a handicap but a requirement, since the BEV homography and the distance thresholds are calibrated for that resolution.

4.3.1 Validity control

The budget sweep includes the original, unbounded-budget image as a reference point. Because the ground-truth graphs were themselves produced from those images, the image arm *must* score 1.000 there. Verifying that it does is what distinguishes a genuine bitrate effect from a defect in the image pipeline.

4.4 Evaluation Metrics

Four metrics are used, defined once and applied identically to both arms.

Delivery rate. Fraction of frames that arrived at all. A property of the channel, not the representation.

Graph-exact rate. Fraction of frames whose reconstruction is node- and edge-identical to the original.

Actor-edge F1. F1 over edges with at least one detected actor at an end. This excludes the road/ego/lane skeleton, which is present regardless of image content and constitutes roughly 38% of a typical edge set.

Safety-relation recall. The fraction of `near_coll` and `super_near` relations involving the ego vehicle that survive. This is the content a braking decision actually reads.

4.5 Summary of the Methodology

The method is to hold the channel, the frames, the detector, the extraction and the scoring constant, and vary exactly one thing — what the transmitted bytes represent. Every claim in Chapter 6 rests on that control.

Chapter 5

Implementation

5.1 Technology Stack

Table 5.1: Technology stack.

Component	Technology	Version
Network simulator	OMNeT++ [12]	6.4.0
VANET framework	Veins [11]	5.3.1
Traffic simulator	SUMO [5]	1.21.0
Physical layer	IEEE 802.11p [4]	6 Mbit/s, 20 mW
Scene graphs	roadscene2vec [6]	—
Object detection	torchvision Faster R-CNN [7]	0.23.0
Deep learning	PyTorch / PyTorch Geometric	2.8.0
Application layer	C++17	—
Tooling	Python	3.12

5.2 The Veins Application

GBSEDAApp extends Veins' DemoBaseAppLayer. The sender loads a queue of payload files, splits each into chunks, Base64-encodes them and broadcasts them as GBSEDApplMessage frames carrying the chunk index, the total chunk count, the file name and the sender's position. The receiver allocates a buffer on the first chunk of a file and writes the re-assembled result to disk.

Several extensions were required.

Configurable timing. The send interval and start time were hard-coded constants, which made the transmittable volume a property of the source code rather than of the

experiment. Both are now NED parameters.

Position stamping and logging. Each message now carries the sender’s coordinates so the receiver can compute the separation at reception. Both nodes append one row per chunk to `tx_log.csv` and `rx_log.csv`, which is what makes delivery analysable against distance.

Partial file delivery. The receiver previously discarded any file that never completed, making a frame missing one chunk indistinguishable from a frame never sent. It now writes incomplete files, zero-filled where chunks are missing — a precondition for the slice-aligned format below.

5.3 The Semantic Tooling

The semantic layer was consolidated into a single importable module, `gbsed_semantic.py`. This matters because the serialisation functions had previously been duplicated verbatim between the pipeline script and an exploratory notebook, an arrangement guaranteed to drift. Two command-line tools wrap it: an encoder that turns a folder of images into payloads plus ground-truth metadata, and a decoder that rebuilds graphs from received bytes and scores them.

5.4 Slice-Aligned Payload Packing

Semantic decompression already tolerates a partial relation set: it writes each delivered slice to its index in T and leaves the remainder zero, and cannot distinguish “this relation was absent from the scene” from “this relation’s slice did not arrive”. The capability existed; the flat serialisation of Layer 3 discarded it, because a single vector cut into fixed-size chunks cannot be parsed once a chunk is missing.

The slice-aligned layout builds the payload as a whole number of blocks of exactly `chunkSize` bytes. Each block begins with a magic number and carries only *complete* relation slices; the first block additionally carries the node block. Because the transport chunks naively at the same size, every network chunk is one self-contained block. A missing block is all zeros, fails the magic check and is skipped.

Slices are ordered safety-first. An earlier ordering that led with the structural `isIn` relation pushed `near_coll` into the second block at small chunk sizes, where losing one chunk cost every safety relation — the exact failure the layout exists to prevent.

The decoder's core required no modification: `decode()` is simply called with whatever index list survived.

5.5 Verification

- The flat serialisation remains the default and still round-trips byte-identically through the published serialiser for all 20 frames.
- The slice-aligned layout produces identical scene graphs to the flat serialisation (0 of 20 differ).
- A drop-one-block test over every block confirms that losing block 0 yields a clean diagnostic error while losing any other block recovers 5/7 relations with safety relations intact.
- The Baseline configuration was reproduced independently on macOS and Linux, returning 14/20 delivered on both.

Chapter 6

Results and Discussion

6.1 Datasets and Experimental Setup

Two sequences of consecutive frames from a lane-change driving dataset were used: seq1 (20 frames) and seq2 (23 frames), at 1280×720 . Scene graphs contain 6–10 nodes, and payloads range from 436 to 1642 bytes with a mean of 811.

The SUMO scenario places two vehicles at a common junction travelling at right angles at 8 m/s, so their separation grows continuously and the sequence crosses the radio range limit mid-run. At this speed both vehicles outlive the send queue, so delivery is governed by distance alone rather than by the sender disappearing. Figure 6.1 shows the geometry and the measured result.

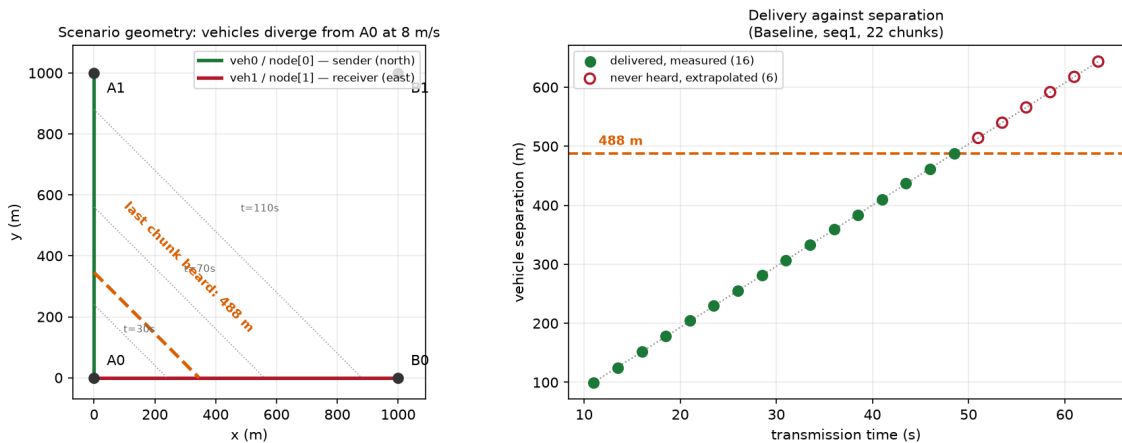


Figure 6.1: Left: scenario geometry. Right: measured delivery against vehicle separation. Separation grows at 10.35 m/s ($r = 0.99999$ against the app’s own logged distances); the last chunk heard was at 488 m.

Five channel conditions were evaluated by varying the receiver noise floor from -98

to -80 dBm.

6.2 Semantic Fidelity over a Real Network

The first result is that the chain survives the port. Across both sequences and all conditions, **every delivered frame reconstructed node- and edge-identically**: actor-edge F1 of 1.000 and safety-relation recall of 1.000 on delivered frames, in every configuration. Figure 6.2 shows a representative frame.

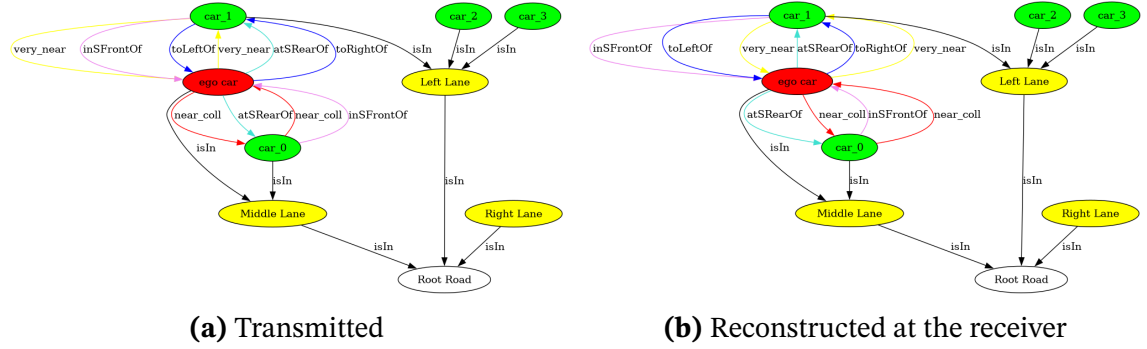


Figure 6.2: A five-actor scene graph before transmission and after reconstruction, delivered at 461 m. Node names, lane structure and all relations are recovered exactly, although names are never transmitted.

This is a stronger statement than the original SNR study can make. Because 802.11p either delivers a frame with a valid checksum or drops it entirely, there is no partially corrupted payload: fidelity is either perfect or the frame is absent. Semantic degradation in a VANET is therefore a question of *which frames arrive*, not of how damaged they are.

6.3 The Fixed-Budget Comparison

Table 6.1 and Figure 6.3 give the central result.

At the GBSSED byte budget, an image is 102×58 to 256×144 pixels at quality 1–5. Across the 14 frames delivered on the best channel the detector found 18 objects in the WebP frames and *zero* in the JPEG frames, against roughly 40 in the originals. Not one of the 32 safety-critical relations survived in either codec.

WebP was given every advantage — it is markedly better than JPEG at very low bitrate, it came in *under* budget (14,604 B against GBSSED’s 16,218), and it held 256×144 where JPEG collapsed to 102×58 . It still recovered nothing.

Table 6.1: Matched-budget comparison. Delivery is identical across arms by construction; only the representation differs.

Arm	Config	Delivered	Graph exact	Safety kept	Actor F1
GBSED	Baseline	70%	70%	32/32	1.000
GBSED	Noise_Low	55%	55%	24/24	1.000
GBSED	Noise_Medium	25%	25%	8/8	1.000
WebP @ budget	Baseline	70%	0%	0/32	0.000
WebP @ budget	Noise_Low	55%	0%	0/24	0.000
JPEG @ budget	Baseline	70%	0%	0/32	0.000
JPEG @ budget	Noise_Low	55%	0%	0/24	0.000

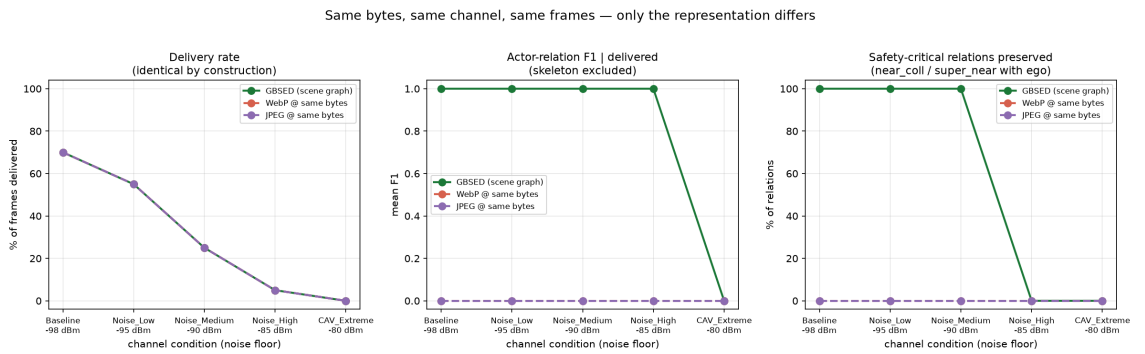


Figure 6.3: Same bytes, same channel, same frames. The three overlapping lines in the left panel confirm the control held. The other two panels show the consequence.

6.3.1 Transmitting the whole frame

For completeness, Table 6.2 reports the naive baseline. The result is not that it is wasteful but that it is impossible: on the best channel the receiver heard 16 chunks, all belonging to frame 0, which requires 1,295 — 1.24% of a single frame.

Table 6.2: The full-image baseline. Payload sizes are also shown in Figure 6.4.

	GBSED	Full JPEG
Total bytes, 20 frames	16,218	24,839,399 (1531×)
Chunks at 1000 B	22	24,847
Simulated time to send	55 s	62,118 s (17.3 hours)
Sender vehicle lifetime	125 s	125 s
Frames delivered (Baseline)	14/20	0/20

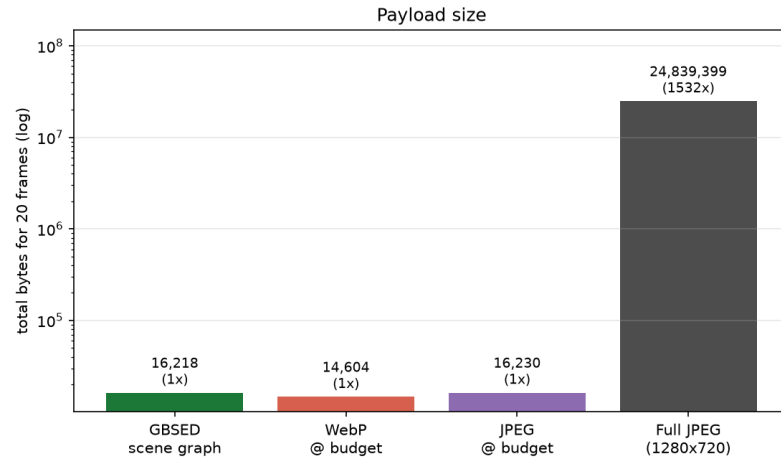


Figure 6.4: Total payload for 20 frames, log scale.

6.4 The Rate–Semantics Curve

The matched-budget result invites the question of how much more bandwidth pixels would need. Figure 6.5 and Table 6.3 answer it.

Expressed as efficiency, GBSED spends 369 bytes per preserved safety relation; WebP at the budget where it finally matches spends 35,368.

The shape of the curves is itself informative. JPEG sits at exactly 0.000 through ×8 and then rises steeply. Image codecs degrade *uniformly*: every object blurs together, and below a threshold the detector fails on all of them at once. That cliff is the mechanism made visible.

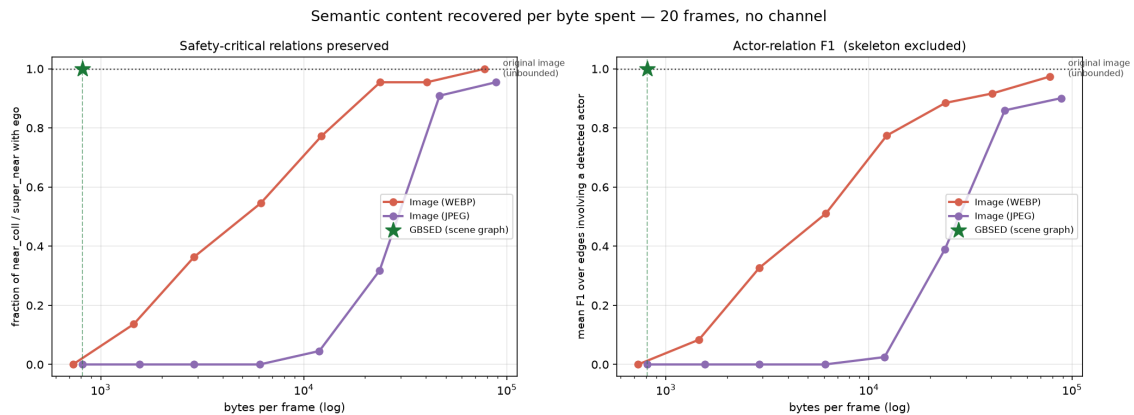


Figure 6.5: Semantic content recovered per byte spent, channel-free. GBSED is the green star at 811 B. The dotted line is the original image at unbounded budget — the arm’s validity control, which it reaches.

Table 6.3: Rate–semantics sweep. WebP first matches GBSED’s safety-relation recall at 77,809 B/frame, i.e. 96×. JPEG does not reach it within the sweep.

Budget	Bytes/frame	WebP safety	WebP F1	JPEG safety	JPEG F1
×1	800	0/44	0.000	0/44	0.000
×2	1,500	6/44	0.084	0/44	0.000
×4	2,900	16/44	0.327	0/44	0.000
×8	6,100	24/44	0.511	0/44	0.000
×16	12,100	34/44	0.774	2/44	0.025
×32	23,700	42/44	0.885	14/44	0.389
×64	43,500	42/44	0.916	40/44	0.860
×128	83,000	44/44	0.974	42/44	0.901
GBSED	811	44/44	1.000	—	—

6.5 Graceful Degradation via Slice-Aligned Packing

Because semantic decompression tolerates a partial relation set, a reconstruction can succeed from only some of a frame’s chunks — but only if the payload is packed to allow it. The flat serialisation cannot: a single lost chunk leaves the concatenated vector unparseable, so the whole frame is lost. The slice-aligned layout packs whole relation slices into independent, self-describing blocks, ordered safety-first, so a lost chunk removes a relation type rather than the frame.

At 500-byte chunks, where frames genuinely span several chunks, this changes the outcome. With identical bytes, identical delivery, and no additional chunks on the wire at 1000 B (one additional at 500 B), safety-relation recall under partial delivery improves as shown in Table 6.4.

Table 6.4: Safety-relation recall under partial delivery at a 500-byte chunk size. Same bytes, same channel, same delivered frames.

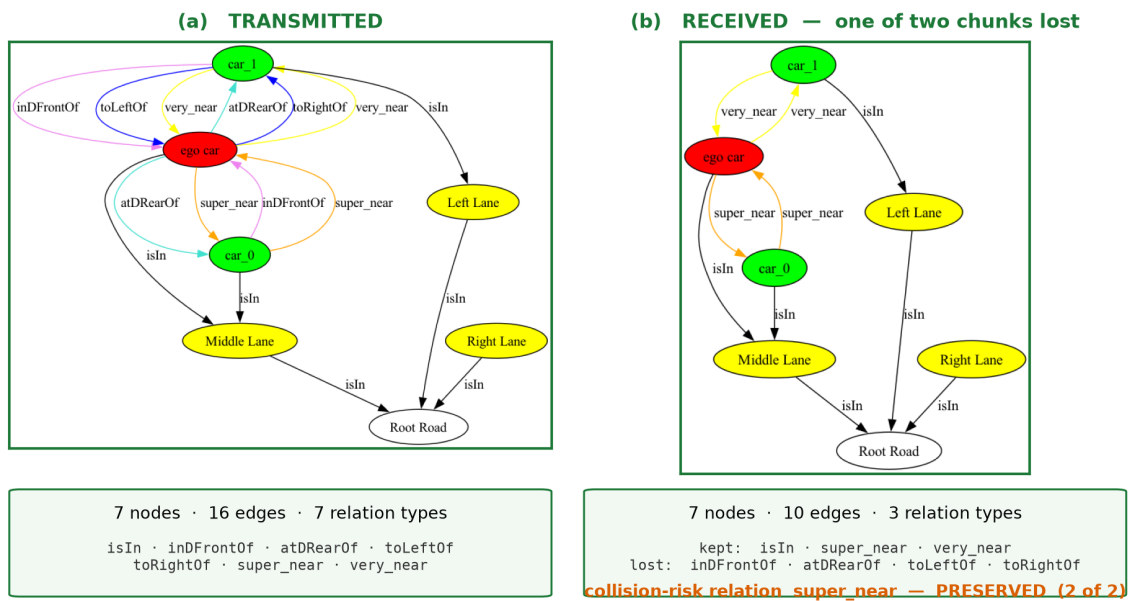
Channel condition	Flat serialisation	Slice-aligned packing
Noise_Low	0.80	1.00
Noise_Medium	0.50	1.00

A single frame illustrates the mechanism (Figure 6.6). Frame 0005 of seq1 occupies 1000 bytes in two 500-byte chunks; chunk 0 was heard at 359 m and chunk 1 never arrived. With the slice-aligned layout the delivered chunk carries the node block together with `super_near`, `isIn` and `very_near`, while the lost chunk carried the four directional relations. The receiver therefore recovers the lane structure and every safety-critical relation, losing only directional context.

Had the same chunk been lost under the flat serialisation, the truncated payload would have been zero-filled, silently emptying whichever adjacency matrices fell in the lost region — here `super_near` itself — and leaving only directional relations of no use to a braking decision. That reconstruction would still contain exactly ten edges and score an identical actor-edge F1 of 0.667, yet preserve none of the safety-critical content. **No aggregate structural metric distinguishes the two outcomes**, which is precisely why safety-relation recall is reported as a separate metric in this work. A second point follows: the flat serialisation not only loses the safety relations but cannot detect that it has, since the zero-filled payload parses without complaint and only a comparison against the sender’s hash reveals the damage.

A scene graph survives packet loss: the safety-critical relation is preserved

One frame, 1000 bytes in two chunks. The first chunk was heard at 359 m; the second never arrived.



The payload carries whole relation slices, collision-risk relations first, so a lost chunk costs directional context — where vehicles are relative to the ego car — while the relations a braking decision depends on ride in the first chunk and survive intact.

Figure 6.6: Partial delivery of a single frame. Left: the transmitted graph. Right: the graph reconstructed after one of the two chunks was lost. The collision-risk relation `super_near` rides in the delivered chunk and survives intact; only directional context is lost.

6.6 Corrected Evaluation Defects

Three defects in the inherited evaluation were identified and corrected. They are reported here because each would have distorted the conclusions.

6.6.1 Three metrics that were one number

In every row of the original results, delivery rate, bit-exact rate and mean edge F1 were identical (Figure 6.7). Two distinct causes:

Bit-exact rate $\equiv$ delivery rate is real, not a bug. IEEE 802.11p delivers a frame with a valid checksum or drops it, so a delivered frame is always bit-exact. The column is worth keeping only as an assertion that should never fire.

Mean edge F1 $\equiv$ delivery rate was a genuine defect. The aggregation filtered lost frames with `if r["edge_f1"]`, but their value is the *string* "0.0", which is truthy in Python. Lost frames entered the mean as zeros, and since every delivered frame scores exactly 1.0, the mean collapsed to the delivery rate.

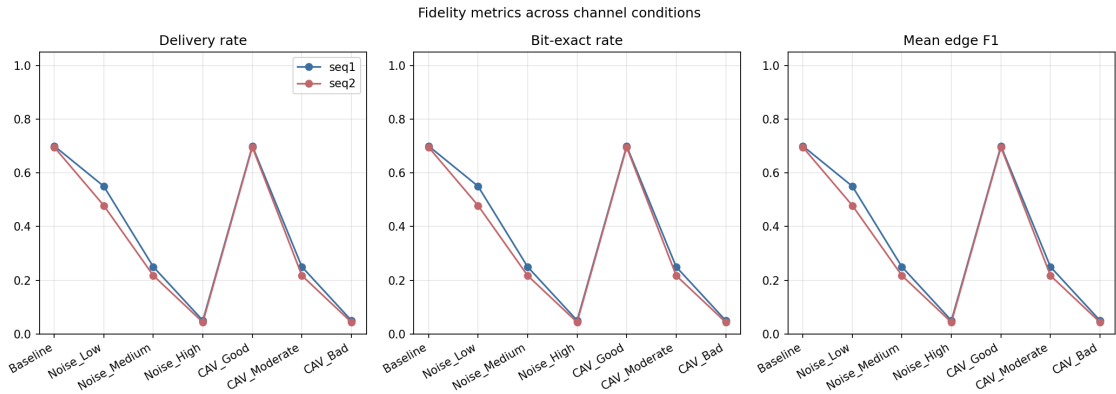


Figure 6.7: The superseded figure, retained as evidence. Three panels plot the identical curve, and the rise at CAV_Good is not a recovery but Baseline replotted under a duplicate configuration.

6.6.2 A metric floor of 0.38

The JPEG arm scored 0.386 mean edge F1 on the best channel *with zero detections*. Every graph contains a road node, an ego node and three lanes joined by `isIn` edges regardless of image content, and that skeleton is roughly 38% of the edge set. Edge F1 therefore ranged over $[0.38, 1]$, with the bottom meaning “understood nothing”. The actor-edge F1 metric introduced in Chapter 4 reads 0.000 for the same input.

6.6.3 Three duplicate channel configurations

Three of the eight configurations differed from three others only by `*.node[1].veinsmobility.x`. That parameter has no effect: node positions come from SUMO over TraCI and are overwritten at every update. The evidence is bit-identical classifier output — `prob_class1` of 0.9753320813179016 for both `Baseline` and `CAV_Good` — which can only arise from identical input graphs. Eight configurations were really five noise floors.

This also biased the accuracy summary, since accuracy was computed over configurations and duplicated conditions were counted twice: `seq1` read $3/7 = 0.43$ where the deduplicated value is $2/4 = 0.50$.

6.7 Interpreting the Results: Why GBSED Works

Three separable mechanisms explain the gap, and they are worth distinguishing because they are usually collapsed into “compression”.

It compresses the right thing. A JPEG spends bits on texture, lighting, road surface and sky, none of which the risk classifier reads. The scene graph spends them on actor identity, position and pairwise relations, which is exactly what it consumes. The 1531× reduction is not a better codec; it is a change in what counts as signal.

It degrades in a different dimension. Image codecs degrade spatially and uniformly, producing the cliff visible in Figure 6.5. A scene graph decomposes into independent relation slices, so bits removed cost relation categories rather than everything at once — as demonstrated in Figure ??.

Semantic content is discrete. A relation is a binary edge; node labels are small integers. These are exactly representable in `float16` and are either right or wrong — there is no “slightly wrong `near_coll`”. Pixels are continuous and every bit removed degrades them. This is why every delivered GBSED frame scores exactly 1.000 and never 0.98.

The honest framing. GBSED is not better compression. It transmits the *output* of perception rather than its *input*, and wins because the receiver’s task needs only that output. The corollary is the limitation: the receiver can never do anything the sender’s extractor did not already encode. A raw image is task-agnostic; a scene graph is committed. The comparison is fair only because the downstream task was fixed in advance.

Chapter 7

Ethical, Social, and Environmental Considerations

7.1 Safety and Accountability

A semantic communication system for vehicles makes a consequential trade. By transmitting an interpretation rather than an observation, it commits to that interpretation being correct. If the sender’s detector misses a pedestrian, no amount of channel reliability will recover them: the receiver has no pixels to re-examine. Our own measurements make this concrete — the graph is recovered perfectly, but it is only ever as good as the perception that produced it.

This has accountability implications. After an incident, a raw video log can be re-analysed with better models; a transmitted scene graph cannot. Any deployment should retain local raw data for forensic purposes even while transmitting only semantics.

7.2 Privacy

Semantic transmission is, incidentally, privacy-preserving. A scene graph contains object categories and coarse positions; it contains no faces, no licence plates and no identifiable imagery. Broadcasting “a car is 4 feet ahead in the middle lane” discloses far less about bystanders than broadcasting the frame that observation came from. This is a genuine ethical advantage of the approach and deserves to be stated as such.

7.3 Environmental and Spectral Sustainability

Radio spectrum is a shared, finite resource. A vehicle transmitting 1.2 MB per frame occupies airtime that every other vehicle in range is denied. Our measurement that a single frame would need 17.3 hours of channel time, while the semantic payload needs 55 seconds for twenty frames, is a statement about spectral efficiency and therefore about how many vehicles a given deployment can support. Reduced transmission also means reduced radio energy consumption per unit of useful information.

7.4 Equity of Access

A system whose bandwidth demands are modest can operate on inexpensive, widely-available radio hardware. A cooperative perception scheme that requires high-bandwidth links would be available only to premium vehicles, concentrating its safety benefits among those who need them least. Semantic communication is comparatively egalitarian in its infrastructure requirements.

Chapter 8

Challenges and Limitations

8.1 Challenges Encountered

Toolchain assembly. SUMO version selection was constrained from three directions at once: Veins 5.3.1 accepts TraCI API versions 15–21, several macOS wheels link against absent libraries, and the OMNeT++ installation was managed by a wrapper that refuses to run outside its own shell. A build-blocking compilation error in upstream Veins also had to be fixed.

Silent failures. Several defects produced plausible wrong answers rather than errors: the truthy-string metric bug, the ineffective mobility parameter, and a malformed XML comment in a route file that surfaced in OMNeT++ as an apparent TraCI version mismatch. Each cost significant debugging time and each is now documented.

Designing a fair comparison. The first instinct — transmit the JPEG — produces an uninterpretable result. Arriving at the matched-budget design required recognising that the byte count had to be controlled per frame.

8.2 Limitations

No negative class. Both evaluated sequences are labelled risky. Without negative examples, classification accuracy cannot distinguish a working model from one that always predicts “risky”, and no false-positive rate is computable. This bounds what the task-level results can claim.

Two sequences. Forty-three frames total is sufficient to establish the mechanisms reported here but not to claim statistical generality.

Two nodes. The scenario has one sender and one receiver with fixed roles. Medium access contention among many vehicles — which is where 802.11p becomes genuinely difficult — is untested.

The slice-aligned format needs a suitable regime. At 1000-byte chunks only 2 of 20 frames span more than one chunk, so the two formats are indistinguishable. The benefit requires either smaller chunks or denser scenes.

Byte reproducibility. Re-encoding on a different platform reproduced 17 of 20 payloads byte-identically; the remainder differ by one unit in the last place of a single float16 feature. The extracted graphs were identical in all 20, so the encoder is reproducible at the level of scene graphs but not of bytes.

No physical-layer comparison. We evaluate 802.11p, the original work evaluates MIMU-OFDM with a CDL channel. The two are complementary; we do not claim one supersedes the other.

Chapter 9

Conclusion and Future Work

9.1 Summary of Key Findings

This project asked whether a graph-based semantic representation is genuinely superior to an image at a fixed bit budget, and answered it under experimental control.

1. The GBSED semantic chain operates losslessly across a realistic VANET. Every delivered frame reconstructs exactly; in 802.11p, semantic degradation is a question of which frames arrive, not how damaged they are.
2. At an identical byte budget over an identical channel, the scene graph preserves 44/44 safety-critical relations while both image codecs preserve 0/44.
3. Images require approximately 96× more bandwidth to convey the same safety-relevant content. GBSED spends 369 bytes per preserved safety relation against WebP’s 35,368.
4. Transmitting whole frames is infeasible rather than merely expensive: 1.24% of one frame arrived within the vehicles’ contact window.
5. Aligning the payload to relation-slice boundaries raises safety-relation recall under partial delivery from 0.50 to 1.00 at no additional airtime cost.

9.2 Contributions

The project contributes a working integration of a published semantic codec with a realistic vehicular network; a controlled experimental design that isolates representational advantage from bandwidth advantage; a rate–semantics curve quantifying the

gap; a payload format that exploits the architecture’s latent robustness; and the correction of three evaluation defects that would otherwise have distorted the reported results.

9.3 Future Directions

Adaptive semantic encoding. The slice-aligned format makes it possible for the transmitter to choose *which* relation slices to send based on live link quality — sending only safety-critical relations when the link is poor. Because the decision is a cheap integer comparison over precomputed slices, this requires no machine learning in the simulation loop.

Multi-vehicle scenarios. Extending beyond two nodes would expose medium access contention and allow study of how semantic payloads share a channel, including whether their small size materially improves aggregate throughput.

A balanced evaluation set. Adding sequences without risk events would permit a genuine accuracy, precision and recall evaluation of the downstream task.

Closing the loop into SUMO. Having the receiving vehicle act on the decoded graph — braking or rerouting — would turn the study from a communication evaluation into a cooperative-perception one.

Integrity protection. The payload carries no checksum, and the flat serialisation parses a truncated payload without complaint. A CRC field in the message would close this cleanly.

9.4 Concluding Remarks

The most useful outcome of this work is not the compression figure but the experimental design that produced it. A compression ratio quoted against raw pixels answers a question nobody asks. Holding the byte budget, the channel, the frames and the scoring constant — and varying only what the bytes mean — produces a claim that is considerably harder to argue with, and one that happens to be far stronger.

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Appendices

Appendix A

Weekly Meeting Summary

Table A.1: Group Contribution and Next Plan Table

Serial	Date	Member 1 Contribution	Member 2 Contribution	Member 3 Contribution	Next Plan

«Fill from your meeting records.»

Appendix B

Code and Implementation Artifacts

B.1 Repositories

- Semantic layer, tooling and experiments:
<https://github.com/sahil0319/gbsed>
- Veins application and simulation scenario:
https://github.com/Loona6/gbsed_veins

B.2 Quantitative Summary: GitHub Metrics

Table B.1: Developer Contribution Metrics. *«Verify against GitHub Insights before submission.»*

Team Member	Total Commits	Additions	Deletions
«Member 1»	« »	« »	« »
«Member 2»	« »	« »	« »
«Member 3»	« »	« »	« »

B.3 Feature Ownership and Technical Contribution

«Assign the features below to the responsible member.»

Team Member	Technical Summary
«Member 1»	Veins application layer. Extended GBSEDAp with configurable send timing, sender position stamping, per-chunk CSV logging, and partial-file delivery. Reworked the SUMO scenario so the two vehicles diverge, converting a trivially lossless link into a measurable range sweep.
«Member 2»	Semantic tooling and experiment harness. Consolidated the duplicated serialisation code into <code>gbsed_semantic.py</code> ; built the encoder and decoder command-line tools, the channel-configuration sweep, and the risk-assessment integration.
«Member 3»	Pixel baseline and evaluation metrics. Designed and implemented the matched-budget image baseline and the rate-semantics sweep; introduced actor-edge F1 and safety-relation recall, and diagnosed the three evaluation defects reported in Chapter 6.

B.4 Verification: GitHub Screenshots



Figure B.1: GitHub commit history.

Image Placeholder: GitHub Contributors Graph Screenshot

Insert screenshot of Insights → Contributors.

Figure B.2: GitHub contributors graph.